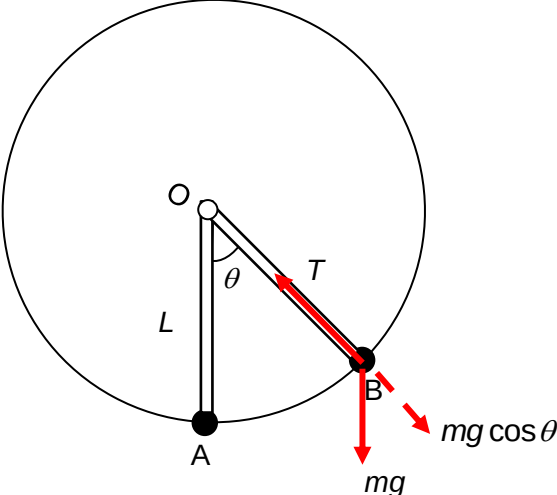
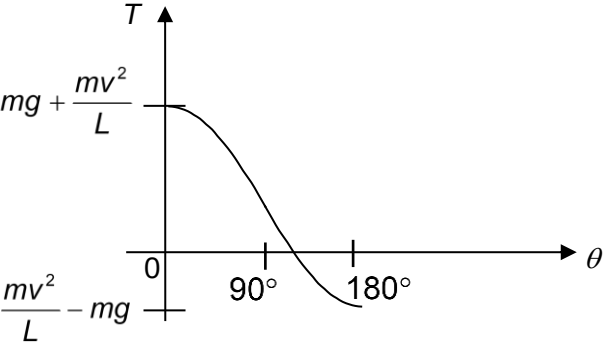


4(a)	Since the direction of motion of the body is always changing, its velocity is not constant and thus a resultant force acts on the body. Since the speed of the body is constant, the resultant force acts perpendicular to its velocity and thus, direction of motion, which is towards the centre of the circle.	[1] [1]
4(b)(i)	 The net force towards centre of the circle provides the centripetal force required to move in a circular path. $T - mg \cos \theta = \frac{mv^2}{r}$ $T = mg \cos \theta + \frac{mv^2}{L}, \text{ as } r = L.$	[1] for statement and equation [1] for $r = L$
4(b)(ii)	The point where the force in the rod changes from tension to compression is when $T = 0$ N. $T = mg \cos \theta + \frac{mv^2}{L} = 0$ $mg \cos \theta = -\frac{mv^2}{L}$ $\cos \theta = -\frac{v^2}{gL} = -\frac{(2.0)^2}{(9.81)(0.80)} = -0.51$ $\theta = 120.6^\circ = 120^\circ$	[1] for $T = 0$ N [1] for correct substitution and answer

4(b)(iii)		[1] graph is cosine and cuts at about 120° [1] correct values
4(b)(iv)	The K.E. of the mass is constant but its G.P.E. is constantly changing. So as the mass is moving upwards, an external device is needed to supply energy to increase its G.P.E. As the mass moves downwards, its G.P.E. is decreasing and the external device must absorb the lost in G.P.E.	[1] energy supply [1] energy absorb
5(a)(i)	A longitudinal wave is one where the oscillations of the wave	[1]